Year 11 Foundation

REVISION PACK

Algebra 8

Name: _____

Class: _____

Algebra 8 · Simultaneous equations (Lesson 12.1)

Worked example: multiplying both equations

Solve the simultaneous equations $3x + 2y = 16$ and $2x + 3y = 14$

Step 1	Make the y the same	$9x + 6y = 48, \quad 4x + 6y = 28$
Step 2	Subtract to remove y	$5x = 20$
Step 3	Divide by 5	$x = 4$
Step 4	Put x in	$3(4) + 2y = 16$
Step 5	Subtract 12, divide by 2	$y = 2$
Step 6	Answer	$x = 4,\ y = 2$

Practice: multiplying both equations

1.1 Solve the simultaneous equations

$3x + 2y = 14$

$2x + 3y = 11$

1.2 Solve the simultaneous equations

$5x + 2y = 9$

$3x + 4y = 11$

1.3 Solve the simultaneous equations

$2x + 5y = 11$

$3x + 2y = 11$

1.4 Solve the simultaneous equations

$4x + 3y = 18$

$3x + 2y = 13$

1.5 Solve the simultaneous equations

$2x + 3y = 14$

$5x + 4y = 21$

1.6 Solve the simultaneous equations

$3x + 4y = 10$

$2x + 5y = 9$

Algebra 8 · Simultaneous equations (Lesson 12.1)

Worked example: different signs

Solve the simultaneous equations $4x + 3y = 10$ and $2x - y = 0$

Step 1	Make the y the same	$2x - y = 0 \xrightarrow{\times 3} 6x - 3y = 0$
Step 2	Different signs: add	$10x = 10$
Step 3	Divide by 10	$x = 1$
Step 4	Put x in	$2(1) - y = 0$
Step 5	Add y to both sides	$y = 2$
Step 6	Answer	$x = 1,\ y = 2$

Practice: different signs

1.7 Solve the simultaneous equations

$3x + 2y = 9$

$4x - y = 1$

1.8 Solve the simultaneous equations

$5x + 3y = 7$

$2x - y = 5$

1.9 Solve the simultaneous equations

$2x + 3y = 9$

$5x - 2y = 13$

1.10 Solve the simultaneous equations

$3x + 2y = 8$

$5x - 4y = 6$

1.11 Solve the simultaneous equations

$4x + y = 6$

$3x - 2y = -1$

1.12 Solve the simultaneous equations

$2x + 5y = 3$

$3x - y = 13$

Algebra 8 · Simultaneous equations (Lesson 12.1)

Worked example: word problems

2 pens and 3 rulers cost 1.300 KD. 3 pens and 1 ruler cost 1.250 KD. Work out the cost of one pen and the cost of one ruler.

Step 1	Use letters	$2p + 3r = 1.3, \quad 3p + r = 1.25$
Step 2	Multiply by 3	$9p + 3r = 3.75$
Step 3	Subtract to remove r	$7p = 2.45$
Step 4	Divide by 7	$p = 0.35$
Step 5	Find r	$r = 1.25 - 3(0.35) = 0.2$
Step 6	Answer	$\text{pen } 0.350 \text{ KD}, \text{ ruler } 0.200 \text{ KD}$

Practice: word problems

1.13 2 wraps and 3 teas cost 3.000 KD. 3 wraps and 1 tea cost 2.750 KD. Work out the cost of one wrap and the cost of one tea.

1.14 4 pens and 1 notebook cost 2.250 KD. 2 pens and 3 notebooks cost 1.750 KD. Work out the cost of one pen and the cost of one notebook.

1.15 3 sandwiches and 2 coffees cost 6.000 KD. 2 sandwiches and 5 coffees cost 6.750 KD. Work out the cost of one sandwich and the cost of one coffee.

1.16 2 cakes and 1 water cost 3.000 KD. 1 cake and 3 waters cost 2.750 KD. Work out the cost of one cake and the cost of one water.

1.17 3 apples and 4 oranges cost 3.000 KD. 2 apples and 1 orange cost 1.500 KD. Work out the cost of one apple and the cost of one orange.

1.18 5 pencils and 2 erasers cost 3.400 KD. 3 pencils and 4 erasers cost 4.000 KD. Work out the cost of one pencil and the cost of one eraser.